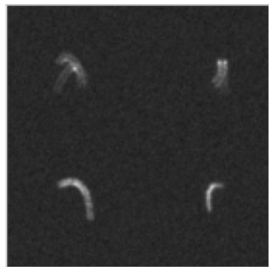


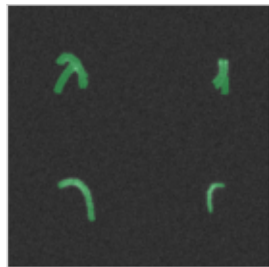
**ENCODE** the image is reduced once

### Input image



sparse fluorescence  
65,536 bytes

### Foreground



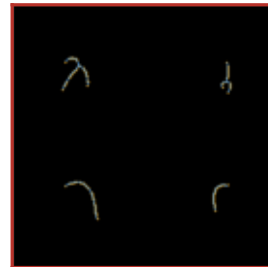
learned cLDice U-Net  
mask IoU 0.90

### Medial axis



skeleton + distance  
transform gives width

### Attributed graph



144 nodes, 141 edges,  
4 components

**THE STORED OBJECT** — the graph itself, 2,077 bytes (31× smaller than the array)

#### per node



position

width

intensity

orientation

× 144 nodes, delta-coded along the graph

#### per edge



connectivity

#### plus



background field

foreground residual

no pixel array is kept; the decoder rebuilds the graph from this stream

### READ THE STRUCTURE directly, no pixels touched

$$\beta_0 = 4$$

connected components, counted on the graph

$$\beta_1 = |E| - |V| + \beta_0$$

cycle rank: path redundancy of the network

#### mean width, edge length

per-edge morphometry in stored units

#### junction degrees

branching pattern of the network

*exact properties of the stored graph: no re-segmentation,  
no re-tracing, identical on every read*

### RENDER THE IMAGE on demand



elliptical PSFs placed at the nodes,  
each aligned to its stored orientation

**PSNR 30.9 dB · FG-PSNR 28.8 dB**

**pixel codec:** bytes → decode → segment → trace → estimate descriptors (re-derived, tool-dependent)

**Nanograph:** bytes → read descriptors · bytes → render image (stored once, then read)